Un algoritmo para encontrar cotas sistólicas a partir de polígonos hiperbólicos

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Joint work with Anita Rojas Rodríguez

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Introduction

Let S be a compact Riemann surface of genus at least 2.

- Systoles: non-homotopically trivial simple closed geodesics of **minimal length**.
- Systolic length (sys): length of any systole.
- Kissing number (Kiss): number of systoles, up to free homotopy.

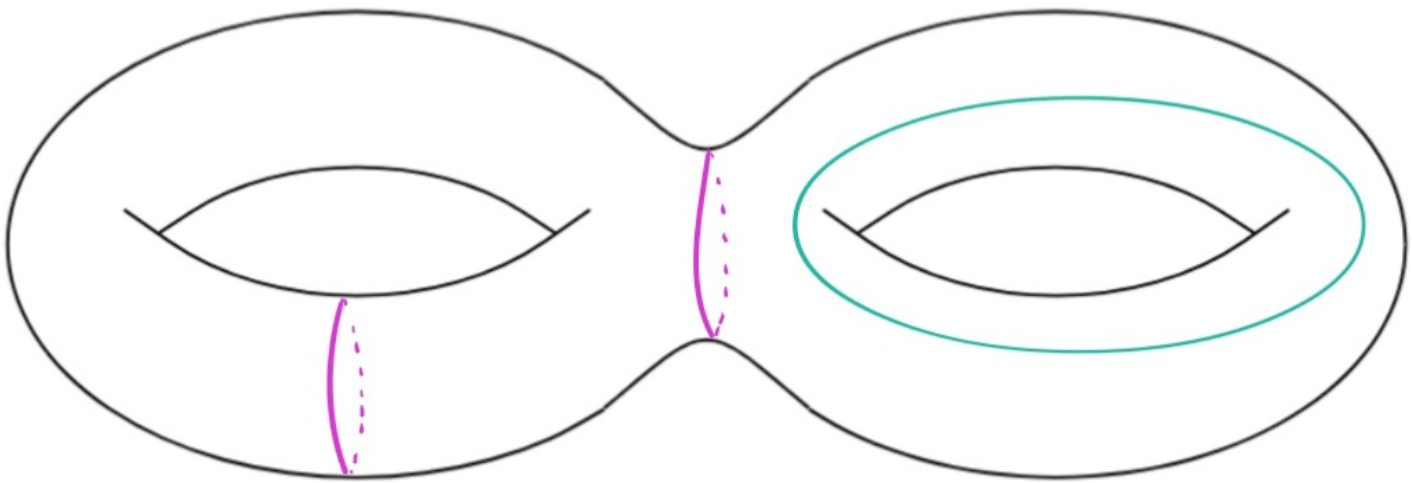


Figure: Pink curves are systoles.

Introduction

Objective: Compute explicit bounds for sys and Kiss.
Strategy: Computational approach.

- Input: abstract finite group and a signature.
- Output: Fundamental polygon for S and candidates to systoles.
- **Reference:** Antonio Behn, Rubí Rodríguez and Anita Rojas (2013).

Guiding example: **Bolza surface** B, endowed with an action of $\mathrm{Aut}(B) \cong \mathrm{GL}_2(3)$ (of order 48) with signature $(0; 8, 3, 2)$.
Affine model: $y^2 = x^5 - x$.

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Group actions and fundamental polygons

Uniformization

- S can be realized as a quotient $\mathbb{D}/\Gamma$, with Γ a Fuchsian group.
- Metric: Restrict hyperbolic metric $g_z = \dfrac{4}{(1-|z|^2)^2} dz d\bar{z}$ to a fundamental domain for Γ.

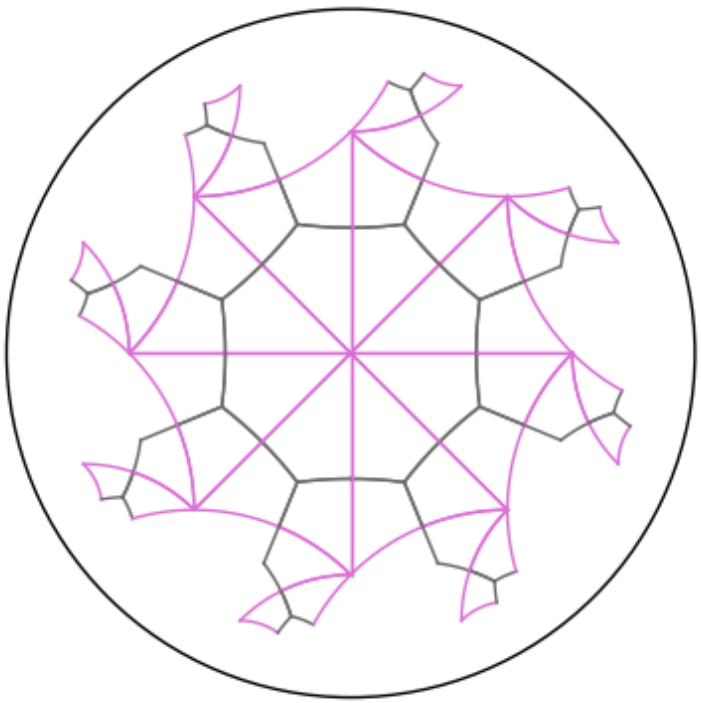


Figure: Fundamental polygon for the Bolza surface.

Closed geodesics

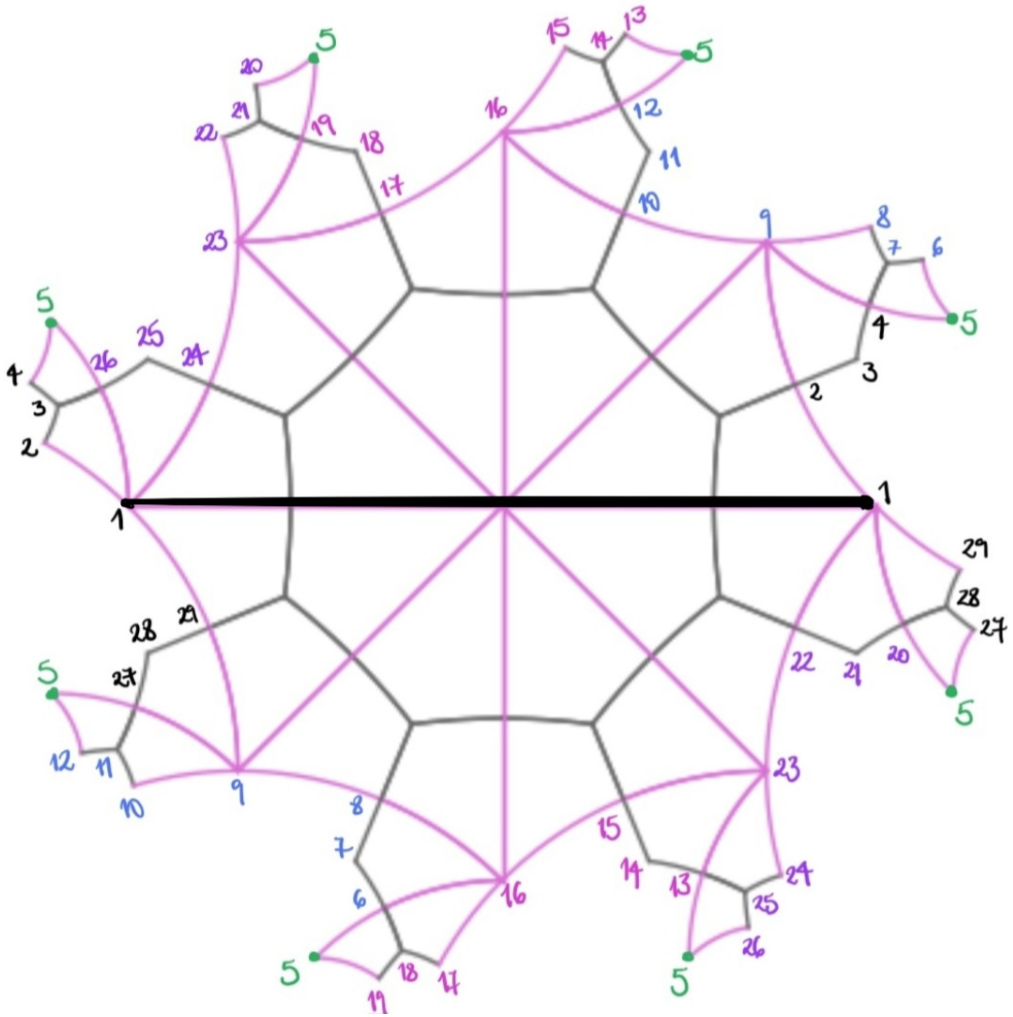


Figure: A closed geodesic on B.

Group actions

- Action of G on S: injective group homomorphism $\varphi\colon G \to \mathrm{Aut}(S)$.
- Surface-kernel epimorphism: group homomorphism $\theta\colon \Delta \to G$ with Δ uniformizing S/G and $\ker(\theta) = \Gamma$.
- Signature: $(0; 8, 3, 2)$. The surface $B/\mathrm{GL}_2(3)$ has genus zero.

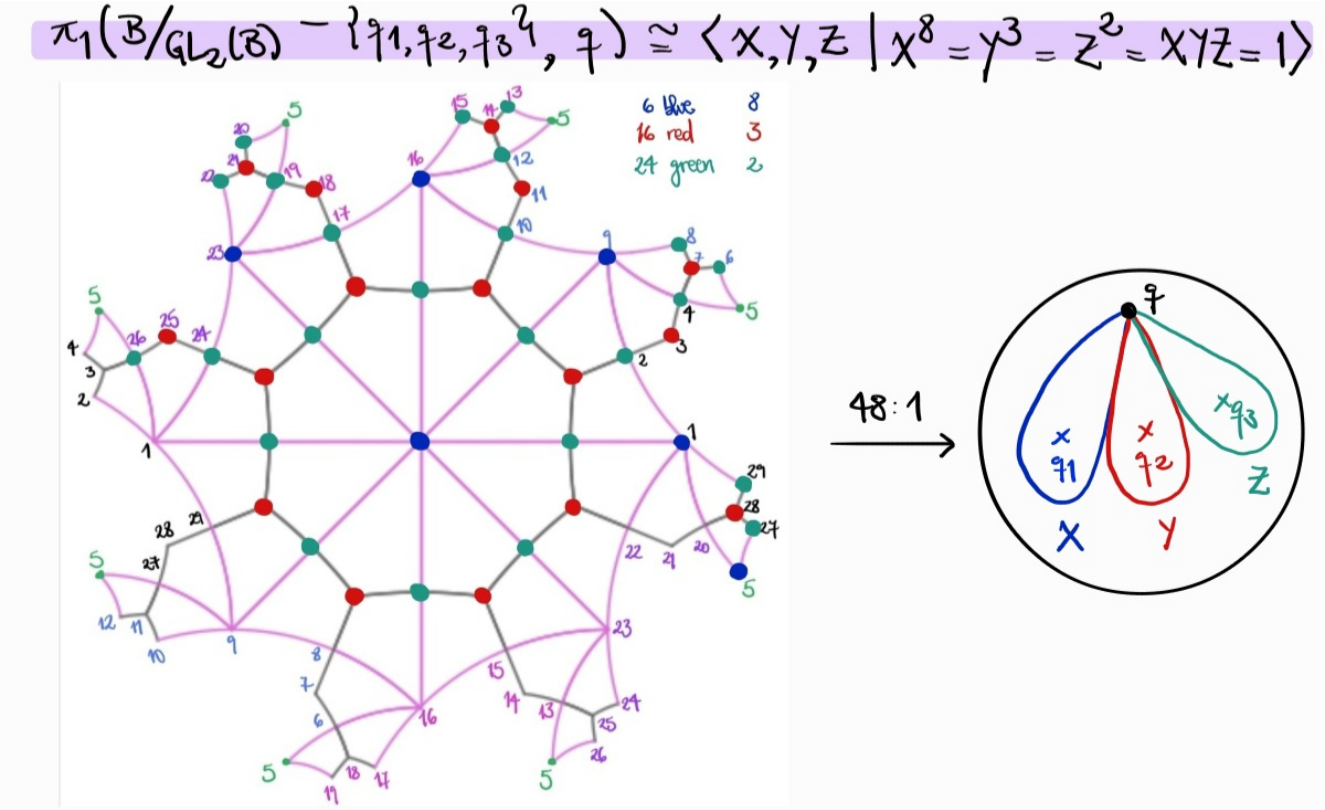


Figure: Fundamental group of the quotient.

Contents

Algorithm
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Uniformizing the quotient

$B/GL_2(3)$ is uniformized by $\Delta \cong \langle X, Y, Z \mid X^8 = Y^3 = Z^2 = XYZ = 1 \rangle$,

$$X = \begin{pmatrix} e^{i2\pi/8} & 0 \\ 0 & 1 \end{pmatrix}$$

$$Y = \begin{pmatrix} 1 & z_0 \\ \overline{z_0} & 1 \end{pmatrix} \begin{pmatrix} e^{i2\pi/3} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -z_0 \\ -\overline{z_0} & 1 \end{pmatrix}$$

$$Z = XY.$$

The point z_0 is constrained by the hyperbolic trigonometric laws.

Fundamental quadrilateral depends on the signature

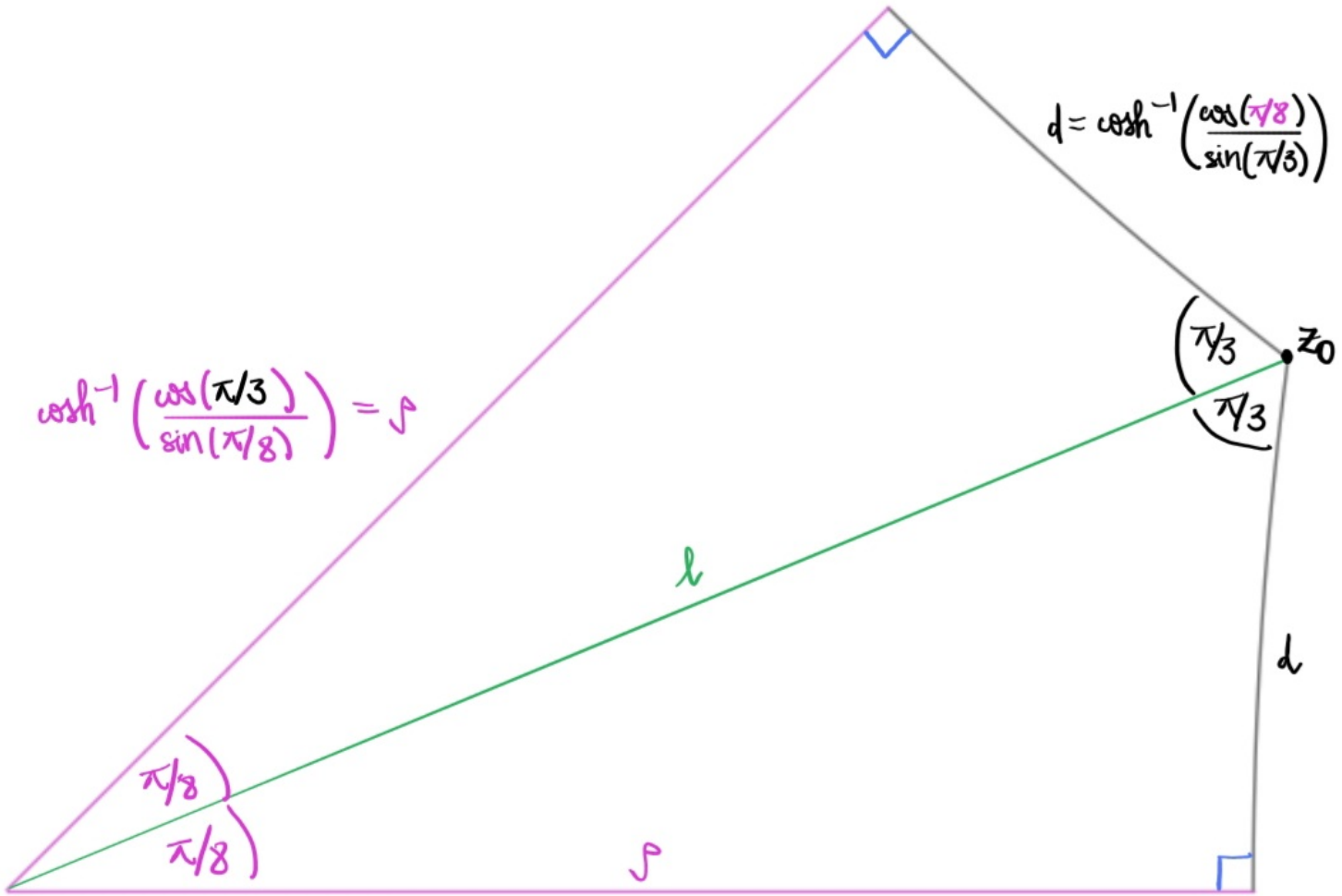


Figure: Fundamental quadrilateral Q for Δ. Its edges are geodesics.

Generating vector

- Compute Γ from an explicit $\theta\colon \Delta \twoheadrightarrow \mathrm{GL}_2(3)$.
- Generating vector: $\theta(X), \theta(Y), \theta(Z) \in \mathrm{GL}_2(3)^3$ of order 8, 3 and 2 respectively.
- **Using GAP interface through Sage:**
  - Presentation:

$$\left\langle x, y, z \mid x^8 = z^2 = yzx = y^3 = (x^{-1}y^{-1}zx^{-1})^2 = zx^{-3}y^{-1}x^4y^{-1} = 1 \right\rangle$$

  - Elements:

$$\left\{ x^j, x^j y, x^j xyx, x^j xyx^2, x^j xyx^3, x^j xyx^3 y \mid 0 \leq j \leq 7 \right\}.$$

Fundamental polygon

Fundamental polygon: Union of non Γ-equivalent copies, $P = \bigcup_{W \in M} W(Q)$,

$$M = \left\{ X^j, X^j Y, X^j XYX, X^j XYX^2, X^j XYX^3, X^j XYX^3 Y \mid 0 \leq j \leq 7 \right\}.$$

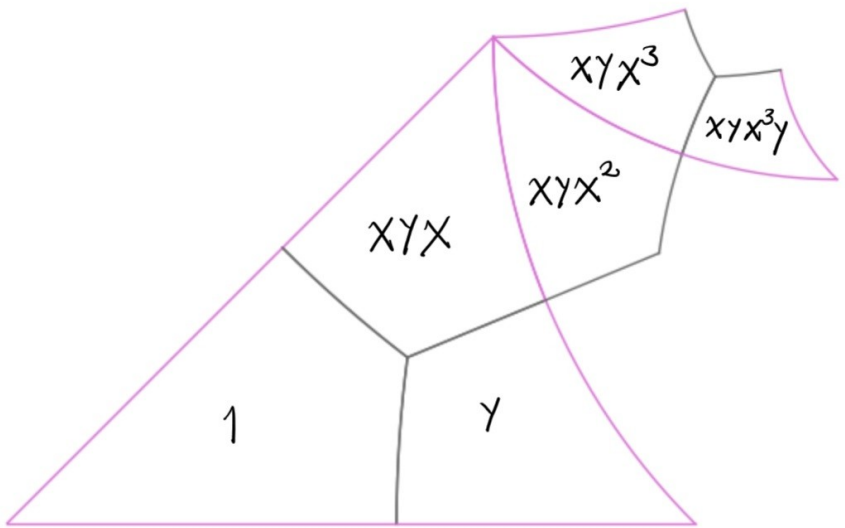


Figure: Eighth part of the polygon ($j = 0$).

Side-pairing (performed by hand)

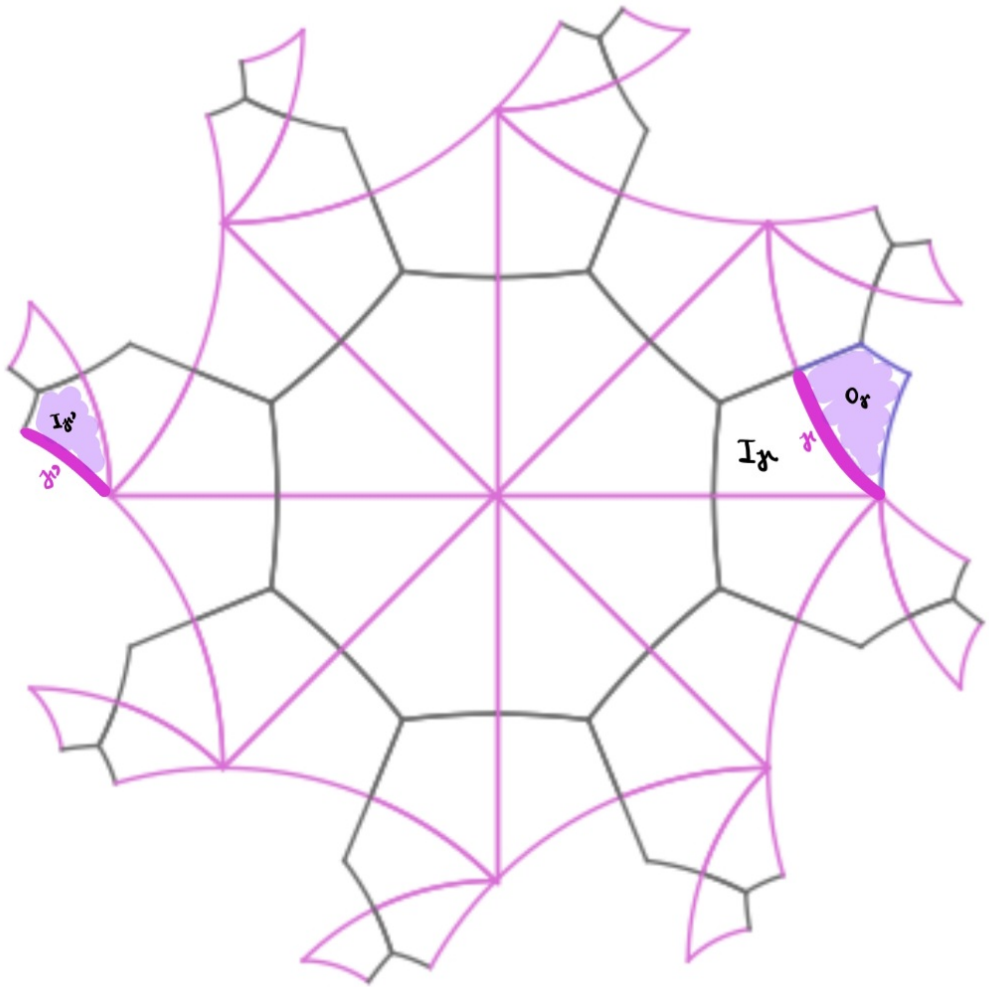


Figure: Γ-identification of boundary curves.

Boundary vertices

The boundary of P has 29 vertices and 32 edges.

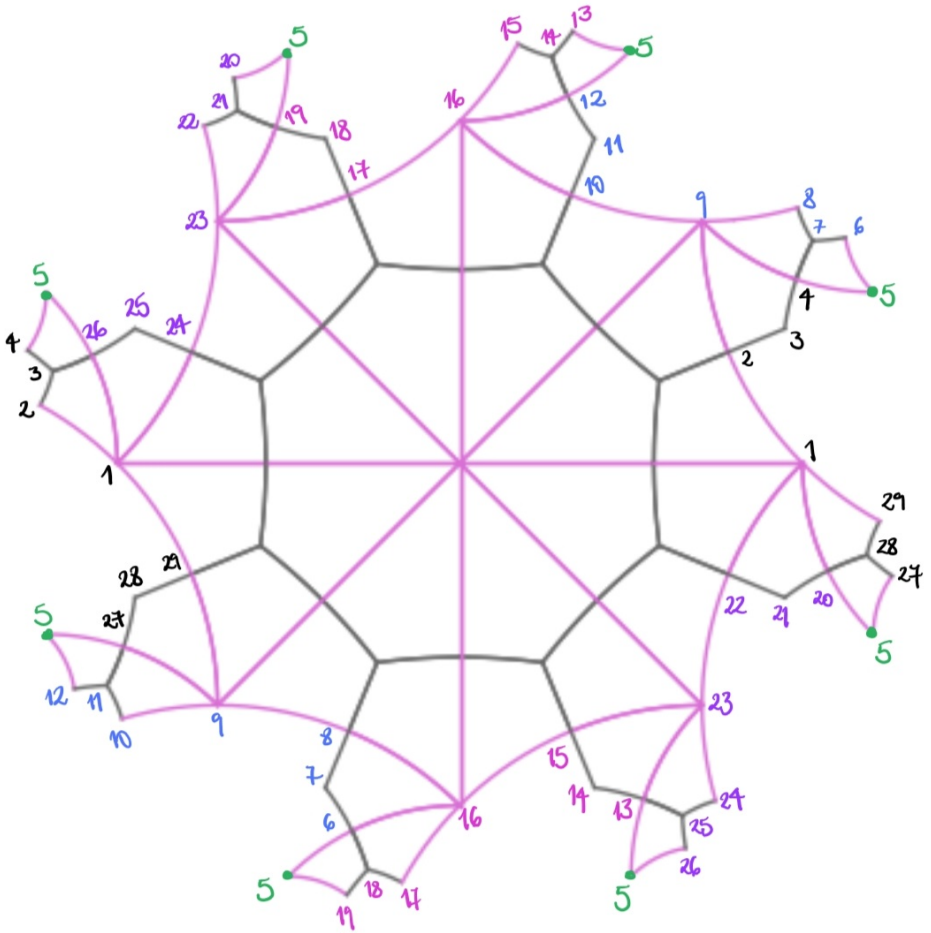


Figure: Γ-identification of the boundary vertices.

Homology basis from relative homology

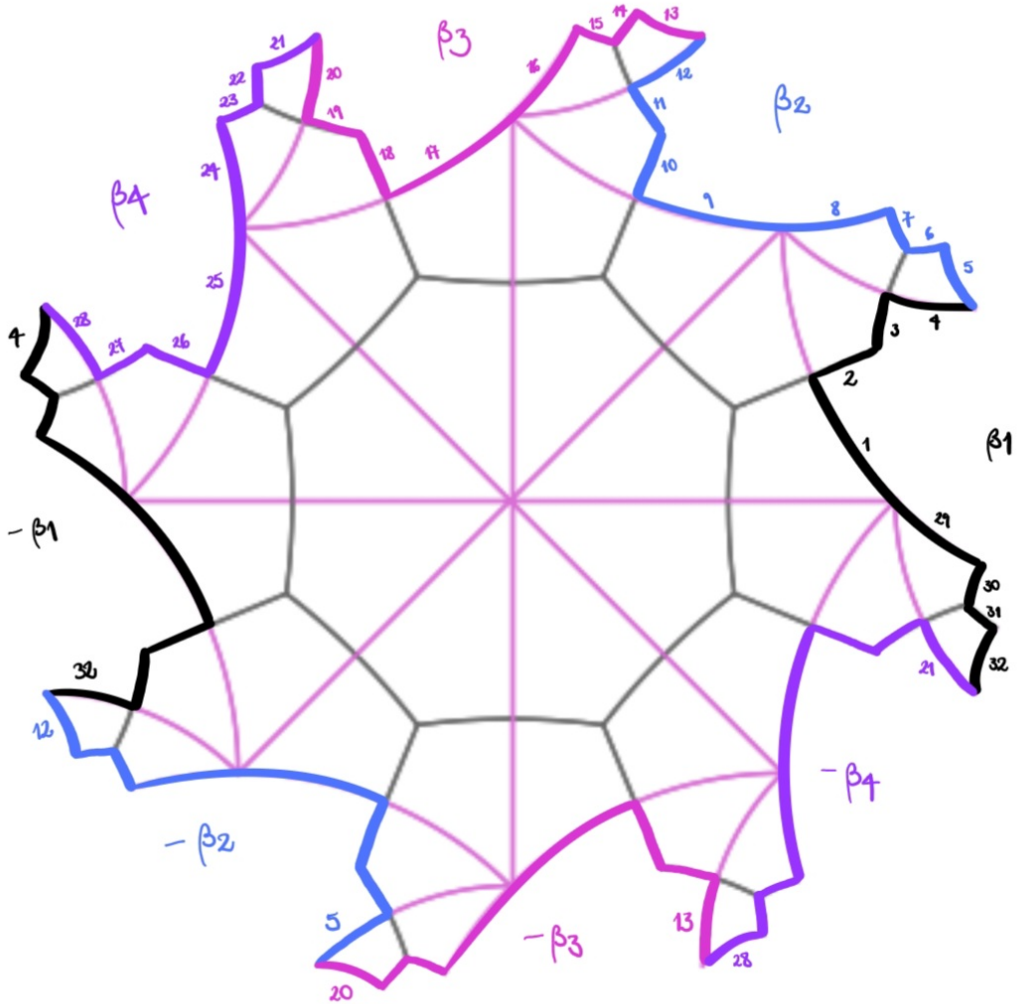


Figure: Homology basis for the Bolza surface.

Geodesic representatives (by visual inspection)

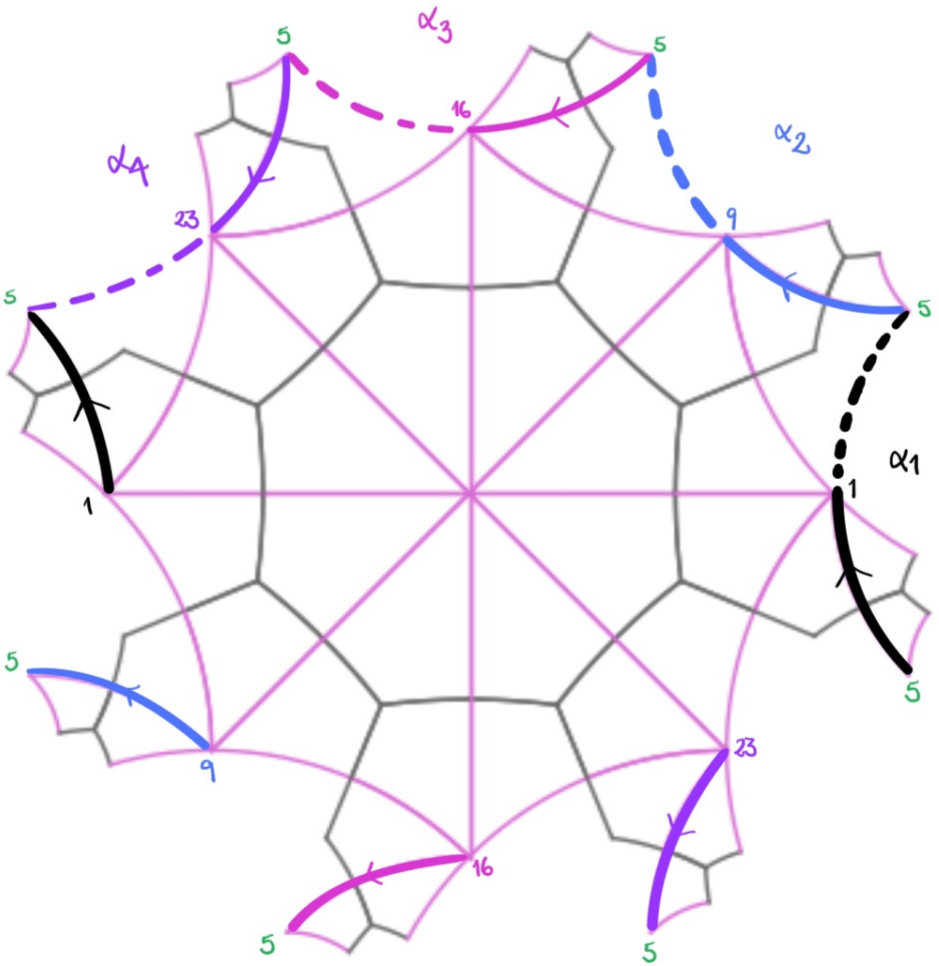


Figure: Homology basis formed by closed geodesics. Each one has length $2\cosh^{-1}(1+\sqrt{2})$.

We found systoles!

- According to Schmutz Schaller (1993), $\mathrm{sys}(B) = 2\cosh^{-1}(1+\sqrt{2}) \approx 3.0571$
- It is the biggest value of sys in $\mathcal{M}_2$.

Length of geodesic representatives provides an upper bound on *sys*

Action by isometries on the polygon

$\mathrm{GL}_2(3)$ acts on the polygon by permuting the copies of Q.
Examples: $x \cdot Y(Q) = XY(Q), y \cdot Y(Q) = XYX(Q)$.

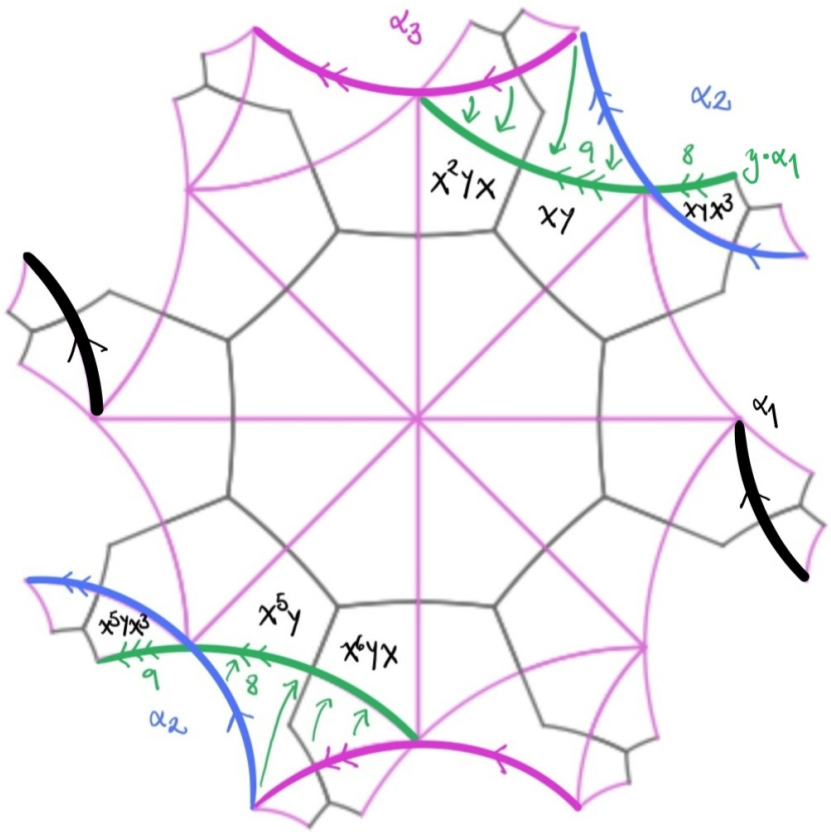


Figure: Image of α_1 under y.

Action in homology

- $\mathrm{GL}_2(3)$ acts by isometries: **orbits of systoles consist of systoles**.
- Orbits' size bound Kiss.
- Representing matrices:

$$\mathrm{HG}_1(x) = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathrm{HG}_1(y) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 \end{pmatrix}.$$

- $\mathrm{Stab}(\alpha_1) = \{1, x^2yx^3, x^6yx^3, x^4\}$. Therefore, B has at least $48/4 = 12$ systoles.

The 12 systoles!

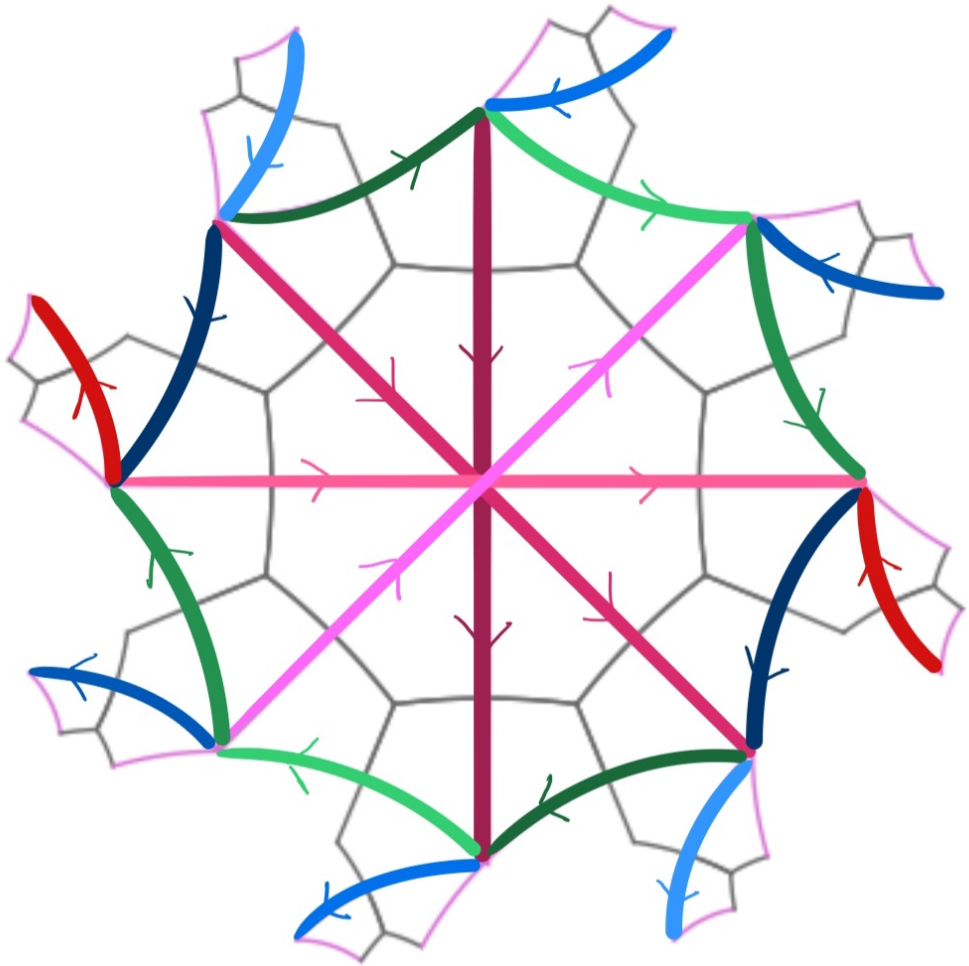


Figure: The 12 systoles of the Bolza surface.

Contents

Application: (Sub)Families on $\mathcal{M}_2$

$\mathcal{M}_2$

- $\mathcal{F}_3$ with D_6-action with signature $(0; 3, 2, 2, 2)$,
- $\mathcal{F}_4$ with D_4-action with signature $(0; 4, 2, 2, 2)$.

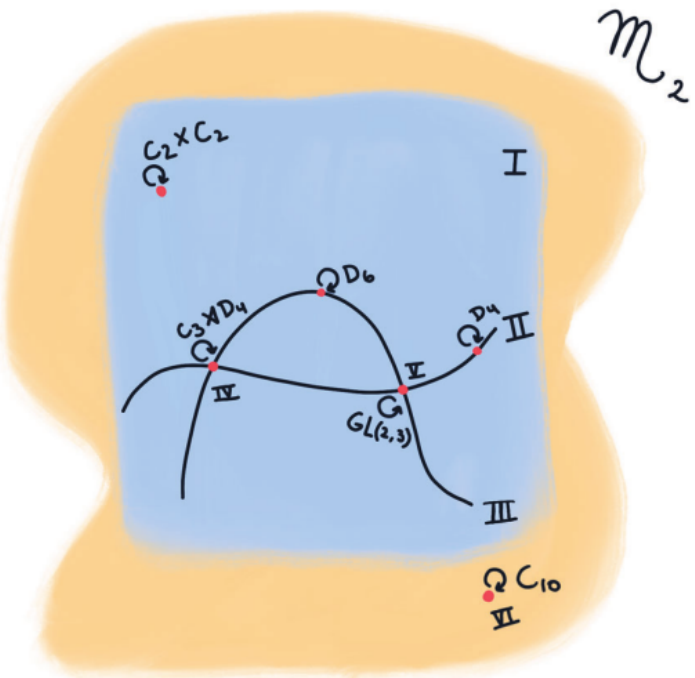


Figure: $\mathcal{M}_2$. Picture from *A Fruitful Interaction Between Algebra, Geometry, and Topology: Varieties Through the Lens of Group Actions*, Rubí Rodríguez and Anita Rojas (2024).

Subfamilies

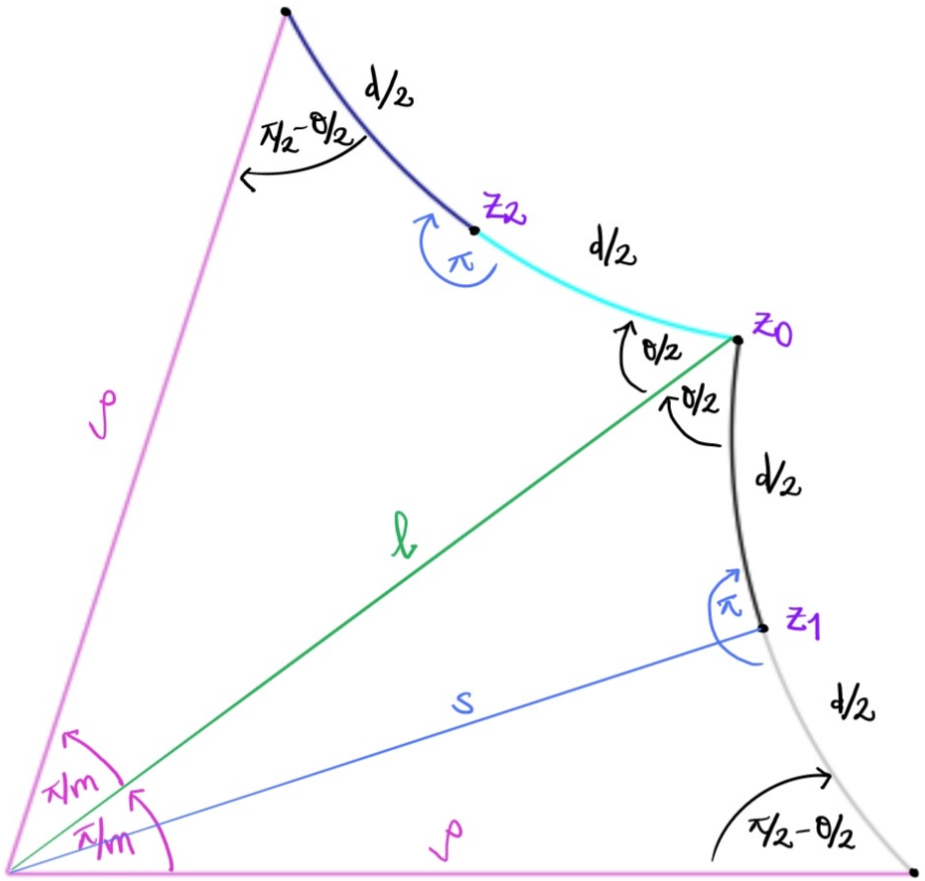
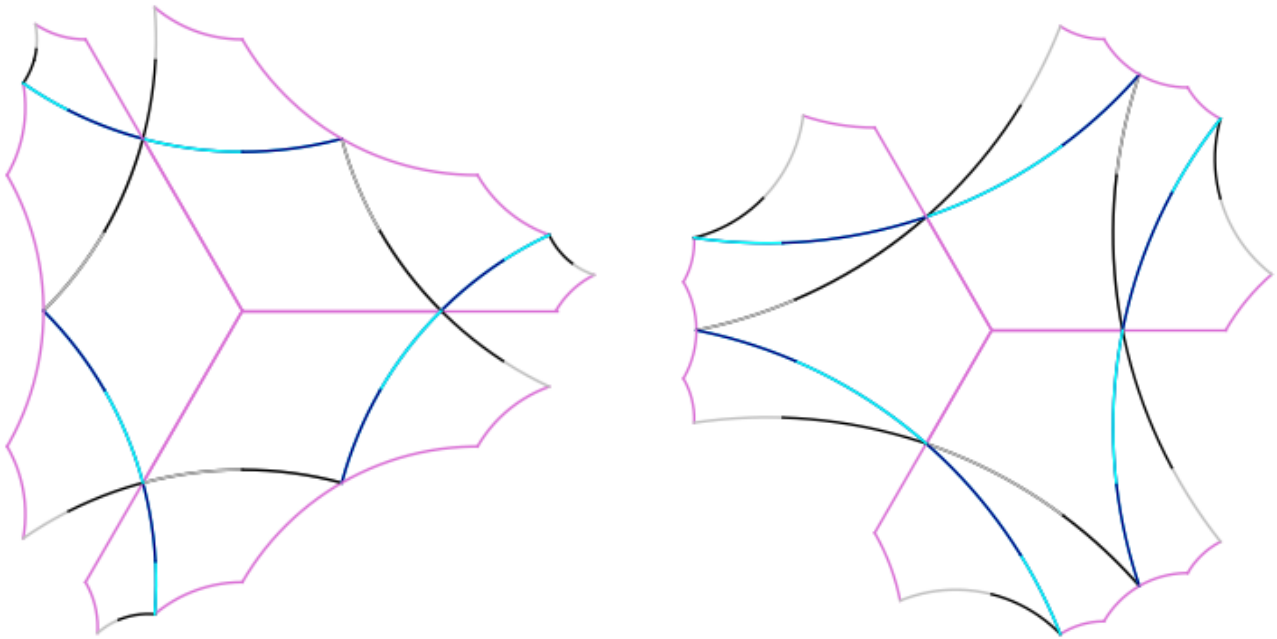


Figure: Fundamental hexagon parametrized by $\theta \in (0, \pi)$.

Fundamental polygons in $\mathcal{F}_3$



(a) $\mathcal{F}_3(\pi/2)$ (b) $\mathcal{F}_3(\pi/8)$

Figure: Members of $\mathcal{F}_3$ with D_6-action.

- Systolic length depends on θ,
- Kissing number only on the group.

Results

GAP ID	Signature	Kiss lower bound	sys upper bound ($\approx$)
(48,29)	(0;8,3,2)	12	3.0571
(12,4)	(0;3,2,2,2)	12	2.6339
(8,3)	(0;4,2,2,2)	8	3.0571

- Bounds for $\mathcal{F}_3$ and $\mathcal{F}_4$ are computed at $\theta = \pi/2$, where its *minimum* is achieved.
- The Bolza surface is contained in both families.

Contents

Conclusions and future work

Summary and beyond

- Construction of fundamental polygons from a group action and a signature.
- Parameter-depending systolic bounds.
- Challenges:
  - Automate steps: side-pairing and how to find geodesic representatives in a given homology class.
  - Complete parameterization of one-dimensional families.

Contact



Figure: QR code to my website: `https://cami-gv.github.io/`

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- Guajardo Vásquez, C. (2025). *On the action of the automorphism group of a compact Riemann surface in its set of systoles*. Master's thesis, University of Chile. Available at ANID's repository `https://repositorio.anid.cl/entities/tesis/3af12d35-b69c-43cd-b95c-6d18aab8c90d`.
- Rodríguez, R. E., and Rojas, A. M (2024). A Fruitful Interaction Between Algebra, Geometry, and Topology: Varieties Through the Lens of Group Actions. Notices of the American Mathematical Society, 71(6).